

During prolonged fasting, which of the following liver enzymes has upregulated activity?

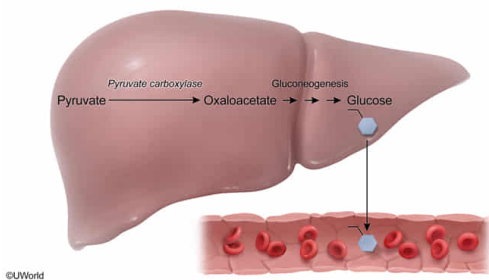
- ☐ A. Glycogen synthase [10%]
- ✓ ☒ B. Pyruvate carboxylase [63%]
- ☐ C. Glucokinase [14%]
- ☐ D. Phosphofructokinase [11%]

Correct

64% answered correctly

Explanation:

Gluconeogenesis is upregulated in the liver during prolonged fasting



The liver is responsible for **maintaining blood glucose levels** during periods of fasting. During fasting, glucose consumption and storage are *downregulated* in the liver and **glucose synthesis is upregulated**. Newly formed glucose is exported into the blood to keep levels constant. In the early stages of fasting, the liver synthesizes glucose by **degrading glycogen** ([glycogenolysis](#)). However, if fasting is prolonged, glycogen stores are depleted, and the liver must upregulate the **synthesis of glucose from smaller precursor molecules** ([gluconeogenesis](#)).

Many of the reactions in gluconeogenesis are glycolysis reactions in reverse, with both processes using the *same* enzymes. However, a few glycolysis reactions are [irreversible](#): phosphorylation of glucose by glucokinase, phosphorylation of fructose 6-phosphate by phosphofructokinase, and phosphate transfer from phosphoenolpyruvate by pyruvate kinase. The enzymes that catalyze these reactions are [tightly regulated](#), and gluconeogenesis uses different enzymes to bypass irreversible steps. During prolonged fasting, the **bypassing enzymes of gluconeogenesis are upregulated**, whereas **key glycolysis enzymes are downregulated**.

The first step in gluconeogenesis is **carboxylation of pyruvate** to form oxaloacetate. This step bypasses the irreversible **pyruvate kinase reaction** and is catalyzed by the enzyme **pyruvate carboxylase**. In the well-fed state, pyruvate carboxylase is *downregulated* so that pyruvate can be used in [other pathways](#), but during prolonged fasting, **pyruvate carboxylase activity is upregulated** in the liver to *increase* the rate of gluconeogenesis.

(Choice A) [Glycogen synthase](#) consumes glucose to produce glycogen. During fasting, the liver must *synthesize* glucose rather than consume it, so glycogen synthase activity is *downregulated*.

(Choices C and D) Glucokinase and phosphofructokinase are tightly regulated enzymes of *glycolysis* that phosphorylate glucose and fructose 6-phosphate, respectively. During prolonged fasting, the activities of these enzymes are *downregulated* in the liver, whereas the gluconeogenesis enzymes—glucose 6-phosphatase and fructose 1,6-bisphosphatase—are upregulated to bypass these irreversible steps and increase gluconeogenesis.

Educational objective:

During periods of fasting, the liver helps maintain blood glucose levels. The liver initially synthesizes glucose by degrading glycogen. However, after glycogen stores are depleted in prolonged fasting, gluconeogenesis is upregulated, increasing the synthesis of glucose from precursors such as pyruvate. In the fasting state, the liver downregulates key enzymes of glycolysis and upregulates enzymes of gluconeogenesis.

Time spent:0 sec**Subject:**
Biochemistry**QID:**402508**System:**
Metabolic Reactions

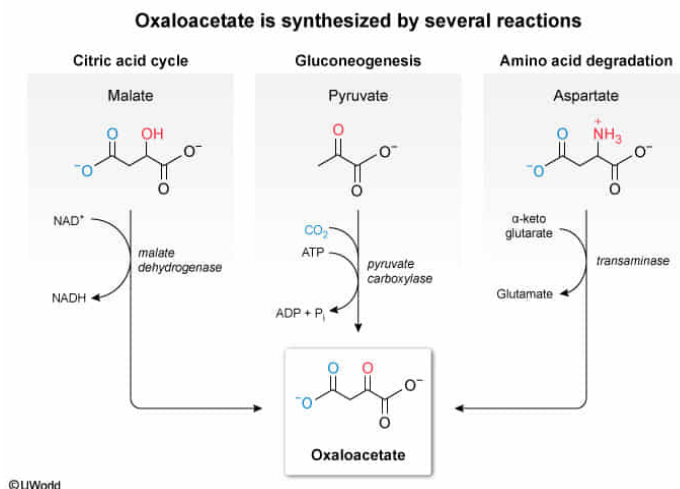
Each of the following molecules can be converted to oxaloacetate by a single enzymatic reaction EXCEPT:

- ☐ A. pyruvate. [19%]
- ☐ B. malate. [5%]
- ✓ ☒ C. alanine. [60%]
- ☐ D. aspartate. [14%]

Correct

60% answered correctly

Explanation:



A **metabolic pathway** is a series of **enzymatic steps** that interconvert metabolic intermediates to produce energy (**catabolism**) or structural molecules (**anabolism**). **Many metabolic pathways intersect** and have an intermediate in common. This intermediate may be synthesized from **multiple precursors**, depending on the pathway used in their synthesis.

Oxaloacetate is an intermediate in several metabolic pathways, including gluconeogenesis, the citric acid cycle, and amino acid degradation. Each uses a different precursor and a different enzyme to catalyze the synthesis of oxaloacetate.

- In the first step of **gluconeogenesis**, the enzyme *pyruvate carboxylase* catalyzes the carboxylation of pyruvate to form oxaloacetate (**Choice A**). In this reaction, CO_2 is added to

pyruvate, and ATP is hydrolyzed to provide the necessary energy.

- In the final step of the **citric acid cycle**, oxaloacetate is synthesized by the oxidation of *malate*, a reaction catalyzed by malate dehydrogenase (**Choice B**). This reaction requires the simultaneous reduction of NAD^+ to NADH. NADH can then enter the electron transport chain and provide electrons necessary for the production of ATP.
- In **amino acid degradation**, amino acids are converted to α -keto acids by **transamination**, in which an amino group is transferred from an amino acid to α -ketoglutarate to synthesize glutamate (**Choice D**). The α -keto acid derivative of *aspartate* is oxaloacetate.

Alanine can be **deaminated** to yield pyruvate, which can *then* be converted to oxaloacetate by pyruvate carboxylase. This process requires *two* enzymatic steps instead of one.

Educational objective:

Metabolic pathways are series of enzymatic reactions that interconvert metabolic intermediates. Many metabolic pathways intersect, and the intermediates at these intersection points may be synthesized from multiple distinct precursors. For example, oxaloacetate is an intermediate in several metabolic pathways: gluconeogenesis, the citric acid cycle, and amino acid degradation.

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Subject:
Biochemistry

QID:402509

System:
Metabolic Reactions

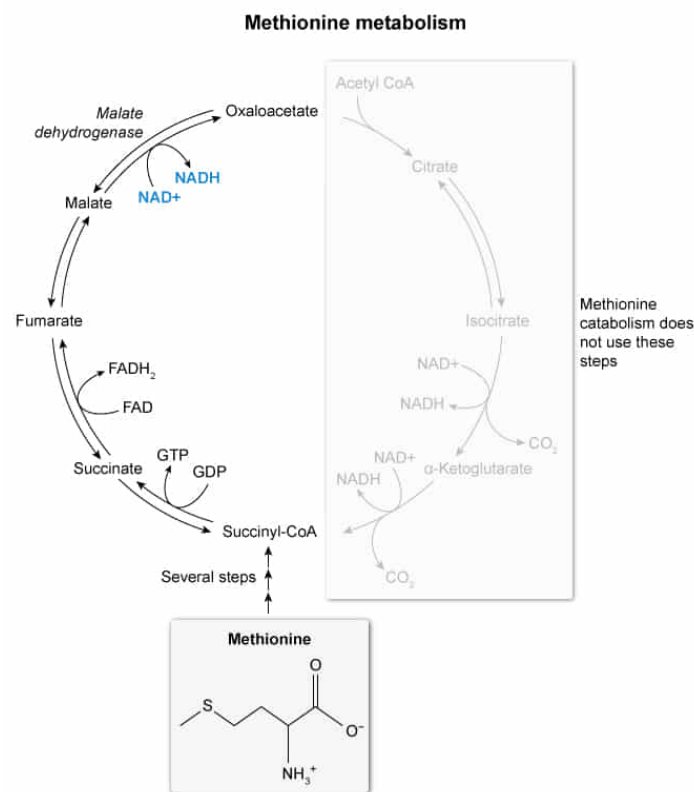
The amino acid methionine can be converted to succinyl-CoA to enter the citric acid cycle. How many NADH molecules can be produced by each methionine molecule that enters the citric acid cycle by this pathway?

- ✔ ☒ A. 1 [55%]
☐ B. 2 [30%]
☐ C. 3 [10%]
☐ D. 4 [2%]

Correct

55% answered correctly

Explanation:



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The **citric acid cycle** is a metabolic process that **oxidizes** a series of molecules to produce the reduced electron carriers NADH and FADH₂, as well as the high-energy molecule GTP. The reduced electron carriers then enter the **electron transport chain** and contribute the electrons necessary for ATP production. Various **molecules can enter the citric acid cycle at different points**, and each produces a **distinct number of reduced electron carriers**, depending on the number of oxidative steps

that remain in the cycle from the point of entry.

According to the question, methionine enters the citric acid cycle by conversion to succinyl-CoA. At this point in the cycle, **two oxidative steps remain**. The first step is the conversion of succinate to fumarate, which produces *one* FADH₂ molecule. The second step is conversion of malate to oxaloacetate, which produces *one* NADH molecule. All other oxidative steps in the cycle require precursors to succinyl-CoA, which do not form in the metabolism of methionine. Based on this scenario, entry of methionine into the citric acid cycle yields **one NADH molecule**.

(Choices B, C, and D) To produce 2, 3, or 4 molecules of NADH, a molecule of α -ketoglutarate, isocitrate, or pyruvate, respectively—or one of their precursors—would be needed. However, methionine is metabolized into succinyl-CoA, which enters the cycle *after* any of these molecules. Furthermore, the oxaloacetate produced at the end of the pathway cannot continue in the citric acid cycle unless *additional acetyl-CoA* is brought in to provide the energy to produce NADH.

Educational objective:

Different molecules can enter the citric acid cycle at different points. The cycle oxidizes these molecules to produce the reduced electron carriers NADH and FADH₂. The number of electron carriers produced depends on the number of oxidative steps remaining in the cycle from the point of entry.

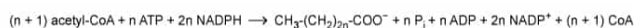
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Subject:
Biochemistry

QID:402510

System:
Metabolic Reactions

Fatty acid synthesis proceeds according to the following reaction:



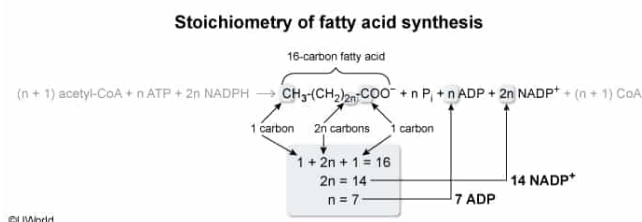
where n is a positive integer. Based on this, how many ADP and NADP^+ molecules will be produced during synthesis of a 16-carbon fatty acid?

- ✓ ☒ A. 7 ADP molecules and 14 NADP^+ molecules [57%]
☐ B. 14 ADP molecules and 28 NADP^+ molecules [7%]
☐ C. 8 ADP molecules and 15 NADP^+ molecules [9%]
☐ D. 16 ADP molecules and 32 NADP^+ molecules [25%]

Correct

57% answered correctly

Explanation:



Fatty acid synthesis produces long hydrocarbon chains that are linked to a carboxylic acid. This process is facilitated by the enzyme **fatty acid synthase**, which **successively links** 2-carbon units (acetyl-CoA) to each other.

The first acetyl-CoA molecule attaches directly to fatty acid synthase. Subsequent acetyl-CoA molecules are **activated** by combining with CO_2 to form malonyl-CoA in a reaction that requires the hydrolysis of 1 ATP molecule to form **1 ADP molecule**. Malonyl-CoA is then linked to the growing fatty acid chain on fatty acid synthase, releasing CO_2 and CoA in the process. A carbonyl carbon in the chain is then **reduced** to the alkyl form in a process that requires oxidation of 2 NADPH molecules to form **2 NADP^+ molecules**. This **process is repeated** until a carbon chain of the desired length is formed.

The question asks how many ADP and NADP^+ molecules will form during synthesis of a 16-carbon fatty acid chain. The reaction scheme shows that for $n + 1$ acetyl-CoA molecules used, n ADP and $2n$ NADP^+ molecules form. The scheme also shows that the resulting **fatty acid contains $2n + 2$ carbons**. If

the total number of carbons in the fatty acid chain ($2n + 2$) is 16, then $2n = 14$ and $n = 7$. Therefore, **7 ADP** molecules and **14 NADP⁺** molecules form in this process.

(Choice B) Fourteen ADP and 28 NADP⁺ molecules would be produced if the fatty acid chain had 30 carbons ($n = 14$, $2n + 2 = 30$).

(Choice C) Eight ADP molecules would be produced in the formation of an 18-carbon fatty acid. In addition, fatty acid synthesis yields an *even* number of NADP⁺ molecules ($2n$), so 15 NADP⁺ molecules cannot be correct.

(Choice D) Sixteen ADP and 32 NADP⁺ molecules would be produced if the fatty acid chain had 34 carbons ($n = 16$, $2n + 2 = 34$).

Educational objective:

Fatty acid synthesis is the process of linking acetyl-CoA units together and reducing carbonyls to the alkyl form. To synthesize a fatty acid containing $2n + 2$ carbons, n ATP molecules and $2n$ NADPH molecules must be consumed, forming ADP and NADP⁺.

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Subject:
Biochemistry

QID:402511

System:
Metabolic Reactions

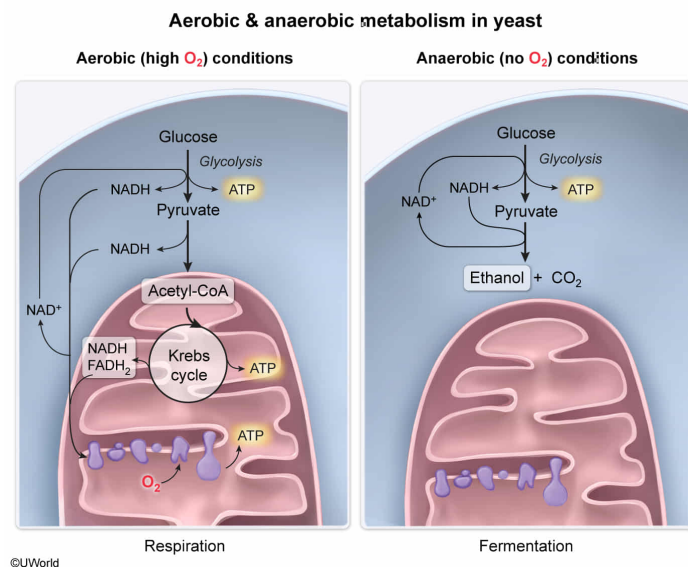
An increase in which of the following would lead to decreased ethanol production in yeast?

- ✓ ☒ A. Partial pressure of oxygen [62%]
☐ B. Pyruvate kinase activity [15%]
☐ C. Concentration of glucose [9%]
☐ D. Krebs cycle inhibitors [12%]

Correct

62% answered correctly

Explanation:



Glycolysis (the breakdown of glucose) **consumes NAD⁺** to produce NADH, and for glycolysis to continue, **NAD⁺ must be regenerated**.

Under **aerobic** (high oxygen) conditions, the NAD⁺ used in glycolysis is regenerated by the **electron transport chain** (ETC) in a process known as respiration. Because oxygen is *required* as the final electron acceptor in the ETC, **anaerobic** (insufficient oxygen) conditions prevents the ETC from functioning and *inhibits* the regeneration of NAD⁺. Accordingly, in anaerobic conditions an alternate process called **fermentation** is necessary to regenerate NAD⁺.

In **higher eukaryotes**, fermentation occurs by reduction of **pyruvate to lactate**. However, in **bacteria and lower eukaryotes** such as yeast

another form of fermentation occurs through decarboxylation of pyruvate to form acetaldehyde, followed by reduction of acetaldehyde to form **ethanol**.

If the partial pressure of oxygen in the environment were *increased*, yeast cells would exhibit *increased* function of the ETC and the need for fermentation to regenerate NAD^+ would *decrease*. Because fermentation in yeast *generates* ethanol, **increased partial pressure of oxygen** (and decreased fermentation) would most likely result in **reduced ethanol production**.

(Choice B) **Pyruvate kinase** catalyzes the transfer of a phosphate group from phosphoenolpyruvate to ADP, forming pyruvate and ATP. If pyruvate kinase activity were increased, *more* pyruvate would be produced. At a given O_2 partial pressure, fermentation of this pyruvate would likely *increase* (not decrease) ethanol production.

(Choice C) The carbon atoms in ethanol originate in glucose. Therefore, increased glucose concentration would make more carbon available and likely increase ethanol production.

(Choice D) The Krebs cycle (ie, citric acid cycle) uses acetyl-CoA, synthesized from pyruvate, to produce NADH and FADH_2 for the ETC. Pyruvate that is converted to acetyl-CoA *cannot* undergo fermentation. **Krebs cycle inhibition** would increase the amount of pyruvate available for fermentation and would most likely increase ethanol production.

Educational objective:

Glycolysis consumes NAD^+ as a substrate. NAD^+ can be regenerated by oxidizing NADH in the electron transport chain in the presence of oxygen, or by fermentation in the absence of oxygen. Fermentation in higher eukaryotes converts pyruvate to lactate, and in bacteria and lower eukaryotes such as yeast it converts pyruvate to ethanol.

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Subject:
Biochemistry

QID:402512

System:
Metabolic Reactions